

KESHAV NARAM

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Professional Summary

Graduate student in Applied Machine Learning with experience developing and deploying end-to-end ML systems in both academic and industry settings. Adept at building real-time speech processing pipelines, training deep learning models, and integrating scalable solutions using Python, PyTorch, and cloud platforms. Strong foundation in algorithms, optimization, and data-driven problem solving. Passionate about applying ML to practical challenges and contributing to impactful engineering.

Skills

- Python
- Flask
- MLflow
- PyTorch
- TensorFlow
- SQL
- Bash
- Docker
- Kubernetes
- Airflow (Apache Airflow and Astro Airflow)
- Jenkins (CI/CD)
- AWS (ECS, CLI)
- Linux
- Git & GitHub

Projects

Hotel Booking Cancellation Predictor (MLOps Pipeline) – Python, Jenkins, AWS, Flask, Airflow Mar 2025 – May 2025

- Built an end-to-end MLOps pipeline to predict hotel booking cancellations using structured data.
- Achieved 99.5% accuracy with a tuned LightGBM classifier; tracked experiments using MLflow and TensorBoard.
- Implemented CI/CD via Jenkins with Pylint, Flake8, Trivy, Docker builds, and DockerHub pushes.
- Orchestrated data workflows and model training using Apache Airflow, ensuring reliable and automated pipeline execution.
- Deployed Flask API on AWS ECS (Fargate) with autoscaling and container health checks using AWS CLI automation.

Depression Detection via Vowel-Like Speech Analysis – PyTorch Aug 2023 – Nov 2023

- Analyzed vowel-like regions in EATD-corpus audio data to distinguish depressed vs. non-depressed speech.
- Used spectral peaks, modulation energy, and excitation features for analysis and visualization.
- Balanced the imbalanced dataset using SMOTE and reshaped audio data for model compatibility.
- Fine-tuned Audio Spectrogram Transformer and ViT on vowel-only mel spectrograms; achieved 87% accuracy and 85% F1-score.

Work Experience

Software Intern – AI/ML [verify] Jul 2022 – Dec 2022

WyzMindz Solutions Pvt. Ltd.

Remote

- Implemented real-time Voice Activity Detection (VAD) to identify speech/silence in audio streams.
- Segmented audio into chunks for live processing and integrated Kaldi/Fairseq-based speech recognition models.
- Trained and fine-tuned Wav2Vec 2.0 models to improve transcript quality from noisy call-center data.
- Built a Flask-based real-time API that combined VAD, speech-to-text, and live transcription in a user-facing interface.

Education

University of Maryland, College Park 2024 - May 2026 (Expected)

Master of Science in Applied Machine Learning (GPA: 3.73 / 4.00)

College Park, MD, USA

Indian Institute of Technology, Dharwad 2020 - 2024

Bachelor of Technology in Electrical Engineering (GPA: 8.08 / 10.00)

Karnataka, India

Courses / Certifications

- DeepLearning.AI: Neural Networks and Deep Learning [verify]
- DeepLearning.AI: Natural Language Processing specialization [verify]

Volunteer Experience

Departmental Academic Mentorship Program (DAMP) [verify] August 2022 - August 2023

Indian Institute of Technology, Dharwad

- Mentored over 20 undergraduate students, guiding them in academic planning, course selection, and career exploration aligned with their interests and goals.
- Developed tailored study plans and resource recommendations, resulting in improved academic performance and clearer career trajectories for mentees.
- Provided ongoing support to students facing academic challenges by identifying root causes and suggesting effective problem-solving strategies.